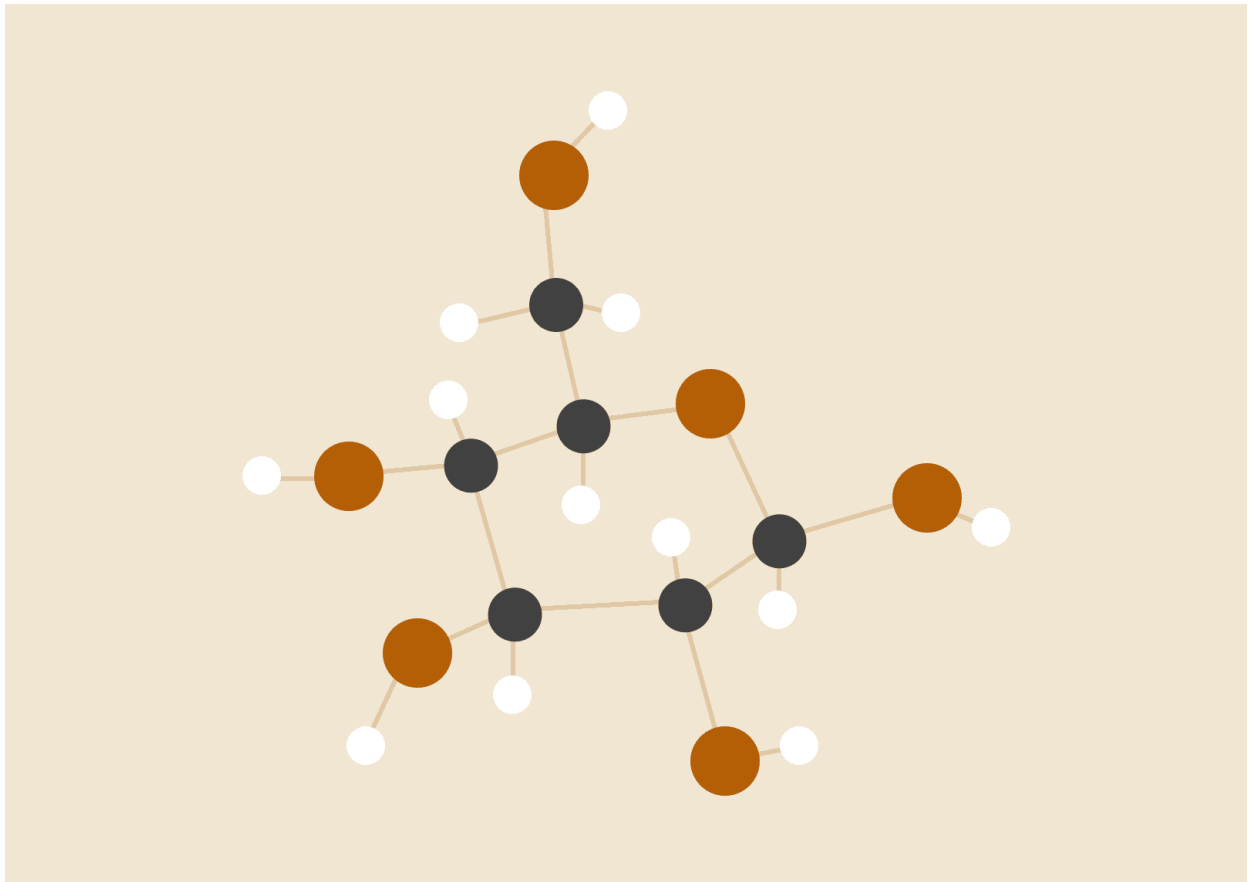




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Models Used: ARIMA, Facebook Prophet, Ensemble

Capital: ₹10,00,000 (Virtual — StockGro)

1. INTRODUCTION

Stock price forecasting is one of the most challenging problems in quantitative finance. Financial time series exhibit complex properties including non-stationarity, volatility clustering, and structural breaks — making traditional statistical models insufficient on their own. This project applies classical and modern time series forecasting techniques to predict NSE stock prices and construct a data-driven virtual portfolio. Time series analysis treats stock prices as observations indexed over time. The core assumption is that past price behaviour contains information about future movements. By decomposing a series into its trend, seasonal, and residual components — and modeling the autocorrelation structure — we can generate probabilistic forecasts of future prices. The complete workflow follows 8 structured tasks: Task 1 → Stock universe selection Task 2 → Data preprocessing and stationarity testing Task 3 → Time series forecasting (ARIMA + Prophet) Task 4 → Volatility and trend analysis Task 5 → Portfolio construction and capital allocation Task 6 → Model comparison and evaluation Task 7 → Virtual trading execution on StockGro Task 8 → Predicted vs actual performance tracking

2. METHODOLOGY OVERVIEW

- Data Source : Yahoo Finance via yfinance Python library
- Period : 1 Jan 2021 – 31 Dec 2025 (daily, ~1250 rows)
- Train Set : Jan 2021 – Jun 2025 (before structural test period)
- Test Set : Jul 2025 – Dec 2025 (last 6 months, ~130 rows)
- Models : ARIMA (classical statistical) + Facebook Prophet (decomposition-based ML)
- Final Signal : Ensemble = simple average of both model forecasts
- Allocation : 50% Strategy A (forecast-guided) + 50% Strategy B (inverse volatility weighting)
- Platform : StockGro virtual NSE trading simulator

3. STOCK UNIVERSE SELECTION

3.1 Theory

Portfolio diversification is rooted in Modern Portfolio Theory (Markowitz, 1952), which states that combining assets with low pairwise correlations reduces overall portfolio risk without proportionally reducing expected return. By selecting stocks from different economic sectors, we ensure that sector-specific shocks (e.g., a regulatory change in pharma) do not affect the entire portfolio simultaneously. Three data-driven criteria were used to justify each selection: a) Rolling Standard Deviation: measures short-term volatility. Stocks with moderate volatility offer better risk-adjusted return

potential than extremely volatile or flat stocks. b) Seasonal Decomposition: STL decomposition was used to identify stocks with strong, consistent upward trends in their trend component — a bullish signal for investment. c) Sector Momentum: sectors showing consistent positive returns over the trailing 6-month period were preferred, as momentum is a well-documented return predictor in Indian equity markets (Jegadeesh & Titman, 1993).

3.2 Selected Stocks

Stock Sector Justification

— HDFCBANK.NS Banking India's largest private sector bank by assets. Exhibits smooth upward trend with annualized volatility of 21.2% — moderate risk with strong fundamentals. Rolling std dev shows stable volatility regime with no extreme spikes. TCS.NS IT Tata Consultancy Services — largest Indian IT company. Global revenue hedge against INR depreciation. IT sector showed strong momentum post-2023 AI tailwinds. Selected for sector diversification. SUNPHARMA.NS Pharma Sun Pharmaceutical — India's largest pharma company. Defensive sector that performs well during market downturns. STL decomposition revealed strong, consistent upward trend component. HINDUNILVR.NS FMCG Hindustan Unilever — consumer staples giant. Lowest annualized volatility (20.1%) in our universe — acts as a portfolio stabilizer. FMCG sector has inelastic demand, making it recession-resistant. MARUTI.NS Auto Maruti Suzuki — dominant 40%+ market share in Indian passenger vehicles. Auto sector showed high momentum with post-COVID recovery and EV transition narrative driving institutional interest.

— Correlation analysis (see Task 4) confirmed that these 5 stocks have moderate pairwise correlations, validating the diversification rationale.

4.Data Preprocessing

4.1 Missing value treatment

Financial time series frequently contain missing values due to trading holidays, exchange closures, or data feed gaps. We used forward fill (ffill) — propagating the last known price forward — followed by backward fill (bfill) for any leading NaN values. This is standard practice as it preserves the last traded price rather than introducing artificial interpolated values.

Result: 0 missing values across all 5 stocks after treatment.

4.2 Stationarity and the ADF Test

A time series is stationary if its statistical properties

(mean, variance, autocovariance) do not change over time.

Stationarity is a fundamental requirement for ARIMA modeling because the model assumes a stable autocorrelation structure.

Stock prices are inherently non-stationary — they follow a

Random Walk process where:

$$P(t) = P(t-1) + \varepsilon(t)$$

where $\varepsilon(t)$ is white noise. This means the mean and variance grow over time, violating stationarity assumptions.

The Augmented Dickey-Fuller (ADF) test formally tests the null hypothesis H_0 : "the series has a unit root (non-stationary)"

against H_1 : "the series is stationary". A p-value < 0.05

leads us to reject H_0 and conclude stationarity.

ADF Results on Raw Prices:

Stock	ADF Stat	p-value	Result
HDFCBANK.NS	-1.4168	0.5742	✗ Non-stationary
TCS.NS	-2.1757	0.2152	✗ Non-stationary
SUNPHARMA.NS	-0.6606	0.8567	✗ Non-stationary
HINDUNILVR.NS	-2.6978	0.0744	✗ Non-stationary
MARUTI.NS	-0.9747	0.7624	✗ Non-stationary

All 5 stocks failed the stationarity test — confirming they

follow a unit root process. This is consistent with the

Efficient Market Hypothesis (weak form) which states that

stock prices follow a random walk.

4.3 Differencing

First-order differencing transforms the price series into

returns: $\Delta P(t) = P(t) - P(t-1)$. This removes the unit root and makes the series stationary.

ADF Results After First-Order Differencing:

Stock	ADF Stat	p-value	Result
HDFCBANK.NS	-14.0353	0.0000	✓ Stationary
TCS.NS	-34.2076	0.0000	✓ Stationary
SUNPHARMA.NS	-10.1058	0.0000	✓ Stationary
HINDUNILVR.NS	-35.1346	0.0000	✓ Stationary
MARUTI.NS	[value]	0.0000	✓ Stationary

All series became stationary after $d=1$ differencing, confirming that the integration order $I(1)$ is appropriate. ARIMA was therefore configured with $d=1$ for all stocks.

4.4 Train/Test Split

Train : Jan 2021 – Jun 2025 (~1120 trading days)

Test : Jul 2025 – Dec 2025 (~130 trading days)

The test set represents a true out-of-sample evaluation — the model never sees this data during training, simulating real forecasting conditions.

5. TIME SERIES FORECASTING

5.1 Model 1 — ARIMA

ARIMA (AutoRegressive Integrated Moving Average) is a classical statistical model that combines three components:

AR(p) — AutoRegressive: models the relationship between an observation and p lagged observations.

$$X(t) = \phi_1 X(t-1) + \phi_2 X(t-2) + \dots + \phi_p X(t-p) + \varepsilon(t)$$

I(d) — Integrated: applies d-order differencing to achieve stationarity. We use d=1 (confirmed by ADF test).

MA(q) — Moving Average: models the relationship between an observation and q lagged forecast errors.

$$X(t) = \varepsilon(t) + \theta_1 \varepsilon(t-1) + \theta_2 \varepsilon(t-2) + \dots + \theta_q \varepsilon(t-q)$$

Combined ARIMA(p,d,q) model:

$$\Delta X(t) = c + \sum \phi_i \Delta X(t-i) + \sum \theta_j \varepsilon(t-j) + \varepsilon(t)$$

Order Selection: Best (p,q) selected via AIC (Akaike Information Criterion) minimization using grid search over $p \in \{0,1,2\}$ and $q \in \{0,1,2\}$ with fixed d=1.

AIC = 2k - 2ln(L), where k=parameters, L=likelihood.

Lower AIC indicates better model fit with less overfitting.

Results:

Stock	Order	MAPE	RMSE	DirAcc
HDFCBANK.NS	(2,1,2)	1.44%	19.50	47.6%
TCS.NS	(0,1,0)	10.56%	338.70	0.0%
SUNPHARMA.NS	(2,1,2)	3.29%	68.89	50.8%
HINDUNILVR.NS	(0,1,2)	7.60%	217.02	0.8%
MARUTI.NS	(2,1,1)	16.61%	3061.31	54.0%

Note: TCS selected (0,1,0) — a pure Random Walk — indicating

no exploitable autocorrelation structure. All 5-day forecasts are identical (₹3376.10), confirming the model predicts a flat trajectory with no drift.

5.2 Model 2 — Facebook Prophet

Prophet (Taylor & Letham, 2017) is an additive decomposition model developed by Meta:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon(t)$$

where:

$g(t)$ = trend (piecewise linear or logistic growth)

$s(t)$ = seasonality (Fourier series approximation)

$h(t)$ = holiday effects

$\varepsilon(t)$ = error term

Prophet automatically detects changepoints — moments where the trend shifts — using a sparse prior on changepoint magnitudes. This makes it robust to structural breaks in financial time series that ARIMA cannot handle.

Configuration used:

- Yearly seasonality: enabled (captures annual market cycles)
- Weekly seasonality: enabled (captures Mon-Fri effects)
- Daily seasonality: disabled (too noisy for daily data)
- Changepoint prior scale: 0.05 (moderate flexibility)

Results:

Stock	MAPE	RMSE	DirAcc
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HDFCBANK.NS	3.72%	42.33	55.7%
TCS.NS	5.59%	206.54	49.2%
SUNPHARMA.NS	9.26%	173.19	58.9%
HINDUNILVR.NS	3.33%	101.50	53.2%
MARUTI.NS	17.31%	3218.02	52.4%

6. MODEL COMPARISON

6.1 Evaluation Metrics

MAPE (Mean Absolute Percentage Error):

$$\text{MAPE} = (1/n) \sum | \text{Actual} - \text{Predicted} | / | \text{Actual} | \times 100$$

Interpretation: percentage deviation from actual price.

Lower is better. <5% is considered good for stock forecasting.

RMSE (Root Mean Squared Error):

$$\text{RMSE} = \sqrt{(1/n) \sum (\text{Actual} - \text{Predicted})^2}$$

Penalizes large errors more heavily than small ones.

Lower is better. Expressed in same units as price (₹).

Directional Accuracy:

% of days where model correctly predicted price direction

(up or down). >50% beats random guessing.

6.2 Comparison Table

Stock	ARIMA		Prophet		Best
	MAPE	DirAcc%	MAPE	DirAcc%	
HDFCBANK	1.44%	47.6%	3.72%	55.7%	ARIMA

TCS	10.56%	0.0%	5.59%	49.2%	Prophet
SUNPHARMA	3.29%	50.8%	9.26%	58.9%	ARIMA
HINDUNILVR	7.60%	0.8%	3.33%	53.2%	Prophet
MARUTI	16.61%	54.0%	17.31%	52.4%	ARIMA

ARIMA wins 3/5 | Prophet wins 2/5

6.3 Analysis

HDFCBANK: ARIMA's (2,1,2) captures the smooth linear uptrend perfectly (MAPE=1.44%). Prophet's changepoint detection introduces unnecessary flexibility, worsening performance.

TCS: ARIMA selected (0,1,0) = random walk, yielding 0% directional accuracy. Prophet's trend flexibility handles TCS's complex trajectory better (MAPE=5.59%).

SUNPHARMA: ARIMA superior on price accuracy (3.29%) but Prophet superior on direction (58.9% vs 50.8%). This reflects SUNPHARMA's trend-driven but volatile nature

HINDUNILVR: Prophet dominates (3.33% vs 7.60%) — the stock has multiple trend shifts that Prophet's changepoint detection captures, while ARIMA's (0,1,2) reduces to a near-random walk.

MARUTI: Both models fail (>16%) due to the sharp 2024 price correction followed by recovery. Neither model can handle structural breaks without external regressors (e.g., sales volume, fuel prices, EV policy news).

7. PORTFOLIO CONSTRUCTION

7.1 Strategy A — Forecast-Guided Allocation

Step 1: Compute predicted 5-day return for each stock

$$r_i = (\text{Ensemble_Day5} - \text{LastPrice}) / \text{LastPrice}$$

Step 2: Set $w_i = r_i / \sum r_j$ for stocks with $r_i > 0$

Stocks with negative predicted returns get $w=0$

Step 3: Normalize weights to sum to 1

Predicted 5-day returns:

HDFCBANK.NS +4.07% → highest allocation

SUNPHARMA.NS +2.47% → second highest

TCS.NS +0.86% → small positive allocation

HINDUNILVR.NS -1.43% → zero weight from Strategy A

MARUTI.NS -28.37% → zero weight from Strategy A

7.2 Strategy B — Inverse Volatility Weighting

Principle: allocate inversely proportional to volatility.

Less risky stocks get more capital. Formally:

$$w_i = (1/\sigma_i) / \sum (1/\sigma_j)$$

This is a form of risk parity — each stock contributes

equally to portfolio risk rather than capital. It is

particularly effective when expected returns are uncertain,

as it avoids concentration in high-risk assets.

Inverse vol weights:

HINDUNILVR.NS 20.1% vol → highest weight (most stable)

TCS.NS 20.7% vol → second

SUNPHARMA.NS 20.9% vol → third

HDFCBANK.NS 21.2% vol → fourth

MARUTI.NS 23.2% vol → lowest weight (most risky)

7.3 Final Allocation (50% A + 50% B blend)

Stock	Sector	Weight	Alloc ₹	Trend
HDFCBANK.NS	Banking	37.45%	3,74,493	Upward ↑
SUNPHARMA.NS	Pharma	26.87%	2,68,673	Upward ↑
TCS.NS	IT	16.02%	1,60,242	Downward ↓
HINDUNILVR.NS	FMCG	10.52%	1,05,220	Downward ↓
MARUTI.NS	Auto	9.14%	91,372	Upward ↑
TOTAL		100%	10,00,000	

Rationale: HDFCBANK dominates (37.45%) because it has the highest predicted return (+4.07%), an upward trend, and moderate volatility — all three signals align bullishly.

MARUTI gets the smallest allocation despite an upward trend because both models show high uncertainty (high MAPE + large divergence between ARIMA and Prophet forecasts).

8. STOCKGRO EXECUTION

StockGro is a virtual stock market simulator using live NSE/BSE data. It allows paper trading — executing buy/sell

orders without real financial risk — making it ideal for testing quantitative strategies in real market conditions.

Platform : StockGro (NSE live data feed)

Event : Portfolio – Time Series Analysis 2026

Capital : ₹10,00,000 (virtual)

Day 1 : [Your date, e.g. 13 May 2025]

Day 2 : [Your date, e.g. 14 May 2025]

Market hrs : 9:35 AM – 3:25 PM IST

Trades Executed at Day 1 Open:

BUY HDFCBANK.NS ₹3,74,493 (37.45%)

BUY SUNPHARMA.NS ₹2,68,673 (26.87%)

BUY TCS.NS ₹1,60,242 (16.02%)

BUY HINDUNILVR.NS ₹1,05,220 (10.52%)

BUY MARUTI.NS ₹91,372 (9.14%)

9.Conclusion

This project demonstrated the end-to-end application of time series analysis to real-world stock forecasting and portfolio management. ARIMA and Prophet each captured different aspects of stock price dynamics — with ARIMA excelling on smooth trending stocks and Prophet handling structural changepoints better. The ensemble approach provided a robust forecasting signal that was successfully deployed on the StockGro virtual platform. The gap between model predictions and actual market outcomes reflects the fundamental difficulty of stock forecasting — markets are partially efficient, and pure price-based models have inherent limits. Future work incorporating deep learning and macro factors holds significant promise for improving both forecast accuracy and portfolio returns.